



OPEN

On the Rayleigh surface wave dynamics of porous piezo-thermoelectric smart material with spatio-temporal nonlocality and fractional higher-order three-phase-lag theory

Mohammed Aldandani¹, Vipin Gupta², Arun Kumar Yadav³, Abhik Sur⁴, Murat Yaylacı⁵ & Soumik Das⁶✉

This study examines Rayleigh-type surface wave propagation in a porous piezo-thermoelectric semi-infinite medium incorporating spatio-temporal nonlocality together with fractional higher-order three-phase-lag heat conduction. The governing equations are formulated within the framework of porosity-dependent thermoelasticity and solved analytically using the wave-mode approach under traction-free boundary conditions. Numerical analysis is carried out to investigate the effects of nonlocal parameters, fractional order, thermal relaxation times, and porosity on phase velocity, attenuation, and particle motion. The results demonstrate that spatio-temporal nonlocal interactions and fractional thermal memory significantly influence wave dispersion and energy dissipation compared to classical thermoelastic models. Several existing theories are recovered as limiting cases, confirming the generality of the formulation. The study provides a useful framework for understanding wave behavior in porous piezo-thermoelectric materials relevant to sensing and surface acoustic wave device applications.

Keywords SDG7, Voids, Piezo-thermo-electric material, KG Nonlocal elasticity, Rayleigh type wave, Fractional order derivative, Three-phase lag model

Surface acoustic waves represent a fundamental mechanism for nondestructive testing, energy transport, and damage evaluation in both engineering and geophysical applications^{1–6}. Surface waves are generally classified into three main types: Rayleigh, Stoneley, and Love waves. Research on Rayleigh waves originates from the foundational study by Lord Rayleigh⁷, who, in 1885, identified their characteristic motion along the traction-free surface of an elastic medium.

Thermoelectric technologies⁸ have also evolved rapidly, with transient modeling of stepped-configuration generators, optimization of non-contact generator parameters^{9–11}. These efforts are complemented by recent demonstrations of stress-wave detection in semiconductor devices^{12,13} and wearable thermoelectric devices inspired by natural systems for enhancing body-heat energy extraction¹⁴. Piezo-thermo-electric materials integrate mechanical, electrical, and thermal field interactions, enabling efficient energy and signal transduction in smart sensors, actuators, surface acoustic wave devices, and MEMS¹⁵. Recent studies have also utilized fractional thermoelastic formulations to investigate anomalous thermal transport and memory-dependent heat conduction phenomena^{16,17}.

¹Mathematics Department, College of Science, Jouf University, P.O. Box: 2014, Sakaka, Saudi Arabia. ²Department of Mathematics, Gurugram University, Gurugram, India. ³School of Basic and Applied Science, K.R. Mangalam University, Gurugram, Haryana, India. ⁴Department of Mathematics, Sister Nivedita University, Kolkata, West Bengal, India. ⁵Department of Civil Engineering, Recep Tayyip Erdogan University, 53100 Rize, Turkey. ⁶School of Physical Science, Amrita Vishwa Vidyapeetham, Mysuru Campus, Mysuru, Karnataka, India. ✉email: soumikdas@my.amrita.edu

Nowacki¹⁸ established coupled thermal elastic diffusion equations, providing deeper insight into the interdependence of mechanical and thermal fields in piezoelectric materials. Cowin and Nunziato¹⁹ formulated linear and nonlinear theories for elastic materials containing voids, while Iovane and Nasedkin²⁰, building upon the Cowin-Nunziato model, developed a linear formulation for piezoelectric media with voids, designed to describe the impact of porosity variation. Recent investigations have also explored advanced computational and thermo-mechanical frameworks involving phase-field modeling, nonlinear vibration control, solar-thermal energy storage systems, and flexible sensing technologies for analyzing complex multi-physics interactions in smart and functional materials^{21–24}.

Edelen and Laws²⁵ made notable contributions to the development of nonlocal interaction theory. However, the applicability of Eringen’s differential model of nonlocal elasticity has been critically examined by Romano et al.²⁶, who presented counterexamples challenging its validity. To address such limitations, Anh and Vinh²⁷ introduced a weakly nonlocal elasticity model in spatial domains and investigated surface wave propagation between two weakly nonlocal media. Later, Anh et al.²⁸ extended this approach to explore Rayleigh type surface waves subject to impedance boundary.

In recent decades, fractional calculus²⁹ has gained significant attention across several scientific and engineering disciplines, including control systems, electromagnetics, quantum mechanics, and nuclear physics. Guo and co-workers³⁰ presented a nonlinear electro-thermo-mechanical analysis of piezoelectric semiconductor junctions exposed to temperature gradients, demonstrating the coupling between thermal loading and electrical response. Saffari et al.³¹ examined the electromechanical energy conversion capability of layered piezoelectric microshell structures under combined moisture and thermal loading, highlighting the role of core geometry in enhancing harvesting efficiency. Several researchers^{32–36} have further explored plane wave propagation in the framework of distinct thermoelastic theories. Gupta et al.^{37–39} analyzed energy ratios in piezo-thermo-electric interface models, considering phase lags and distinct fractional and memory-dependent theories.

Li and his collaborators^{40–43} have extensively investigated coupled thermo-electro-mechanical responses in piezoelectric composites, including stress-gradient thermoelasticity, size-dependent nanostructures, fractional heat conduction, and space–time nonlocal multi-physics modeling. Fekry⁴⁴ analyzed thermo-mechanical stress development in fiber-reinforced porous viscoelastic materials within a nonlocal elasticity setting, emphasizing the interaction between porosity and time-dependent material behavior. Nandi and Pal Sarkar⁴⁵ proposed an advanced fractional thermoelastic framework to investigate heat transfer and stress evolution in porous continua, incorporating multi-phase thermal lag effects.

A comprehensive survey of the existing literature indicates that most investigations on piezo-thermo-electric media are primarily developed within the framework of classical or local elasticity theories, often neglecting the combined influence of spatio–temporal nonlocality, fractional heat conduction, and material porosity. Although some studies have addressed nonlocal elastic effects or fractional thermal responses individually, a coupled formulation incorporating these mechanisms simultaneously remains largely unexplored.

The present study addresses this gap by developing a unified model for Rayleigh-type surface wave propagation in a porous piezo-thermoelectric medium that integrates spatio–temporal nonlocal interactions with fractional higher-order three-phase-lag heat conduction. The formulation accounts for microstructural length-scale effects, thermal memory behavior, and porosity-dependent electromechanical coupling within a single analytical framework. Such interactions become particularly important in micro/nano-structured smart materials, porous piezoelectric composites, and thermally sensitive sensing platforms where classical thermoelastic descriptions are insufficient. Moreover, wave propagation analyses in porous piezo-thermoelectric materials remain relatively limited in the literature, especially under combined nonlocal and fractional thermoelastic environments. By incorporating these coupled mechanisms, the proposed framework provides an extended basis for examining dispersion, attenuation, and energy transport characteristics in advanced porous piezoelectric materials relevant to sensing technologies, surface acoustic wave devices, and nondestructive evaluation applications.

Nomenclature

$\psi =$	Volume fraction of matrix pore	$\tau_i =$	Pyroelectric constants
$\varphi =$	Electrical potential	$\eta_{ijr}, \varepsilon_{ij} =$	Piezo-thermal moduli tensors
$\xi =$	Spatial nonlocality parameter	$\sigma_i =$	Nonlocal equilibrated stress
$C_E =$	Specific heat at constant strain	$D_i =$	Component of dielectric displacement
$\chi =$	Equilibrated inertia of matrix pore	$l_1^* =$	Extrinsic equilibrated body force
$\theta =$	Thermal temperature	$\tau_v =$	Phase lag of thermal disp. Gradient
$E_i =$	Electric field density	$\sigma_{ij} =$	Components of the nonlocal stress
$\tau_q =$	Phase lag of heat flux	$S^* =$	Entropy
$K_{ij} =$	Components of the thermal conductivity	$\beta_{ij}^T =$	Thermal moduli tensors
$\tau =$	Temporal nonlocality parameter	$e_{ij} =$	Component of strain
$c_{ijrl} =$	Elastic stiffness tensor	$\rho =$	Density
$\zeta =$	Nonlocal intrinsic equilibrated body	$K_{ij}^* =$	Components of the rate of thermal force conductivity
$\tau_T =$	Phase lag of the temperature gradient	$q_i^* =$	Component of heat flux
where $h_{ij}, a_1, A_{ik}, \gamma_{ij}, g_i,$ and ν are the constitutive constants that characterize the material characteristics linked with variations in porosity.			

Fundamental equations

Let V denote the domain occupied by an unbounded elastic medium acted upon by body forces F_i . The balance of linear momentum reads

$$\sigma_{ij,j}(\mathbf{x},t) + F_i(\mathbf{x},t) = \rho \ddot{u}_i(\mathbf{x},t), \quad (1)$$

where σ_{ij} denotes the stress tensor incorporating nonlocal effects, u_i is the displacement vector, and ρ is the material density.

Adopting a spatio-temporal nonlocal constitutive law, the stress is given by the convolution^{46,47}

$$\sigma_{ij}(\mathbf{x},t) = \int_{-\infty}^t \int_V C_{ijkl}(\mathbf{x},t;\mathbf{x}',t') e_{kl}(\mathbf{x}',t') dV' dt', \quad (2)$$

where C_{ijkl} is the fourth-order nonlocal elasticity tensor, $\mathbf{x}, \mathbf{x}' \in V \subseteq \mathbb{R}^3$ and

$$e_{kl} = \frac{1}{2} (u_{k,l} + u_{l,k}).$$

denotes the strain tensor.

Substituting (2) into (1) yields the integral form of the nonlocal equation of motion

$$\frac{\partial}{\partial x_j} \int_{-\infty}^t \int_V e_{kl}(\mathbf{x}',t') C_{ijkl}(\mathbf{x},t;\mathbf{x}',t') dV' dt' + F_i(\mathbf{x},t) = \rho \ddot{u}_i(\mathbf{x},t). \quad (3)$$

For practical modeling, the fourth-order kernel is decomposed into the product of a classical elastic tensor and a scalar attenuation function as

$$C_{ijkl}(\mathbf{x} - \mathbf{x}', t - t') = \alpha(\mathbf{x} - \mathbf{x}', t - t') C_{ijkl}^L, \quad (4)$$

where C_{ijkl}^L is the standard local elasticity tensor and α characterizes the spatial and temporal decay of nonlocal interactions.

Using (4), the nonlocal stress is obtained by convolving the kernel function with the classical Cauchy stress

$$\sigma_{ij}(\mathbf{x},t) = \int_{-\infty}^t \int_V C_{ijkl}(\mathbf{x},t;\mathbf{x}',t') e_{kl}(\mathbf{x}',t') dV' dt' = \alpha(\mathbf{x},t) * \sigma_{ij}^L(\mathbf{x},t),$$

where the symbol $*$ denotes convolution carried out jointly over the spatial coordinates and the time variable.

Consequently, in the case of a homogeneous isotropic material, the nonlocal stress field σ_{ij} admits the following representation:

$$\sigma_{ij}(\mathbf{x},t) = \int_{-\infty}^t \int_V \alpha(|\mathbf{x} - \mathbf{x}', t - t'|) \sigma_{ij}^L(\mathbf{x}',t') dV' dt', \quad (5)$$

where $\sigma_{ij}^L = \lambda e_{kk} \delta_{ij} + 2\mu e_{ij}$, is the Cauchy stress tensor, λ and μ are the Lamé constants. Eringen highlighted the physical relevance of the scalar attenuation function by showing that it can effectively capture dispersion characteristics observed in lattice dynamics. The scalar kernel can be represented using a Green's function approach involving a linear differential operator L .

For an infinite medium, the nonlocal kernel operator L can be generally formulated as

$$L\alpha(|\mathbf{x} - \mathbf{x}', t - t'|) = \delta(\mathbf{x} - \mathbf{x}') \delta(t - t'), \quad (6)$$

where $\delta(\cdot)$ denotes the Dirac delta distribution. By operating $L \equiv 1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}$ on Eq. (5) with the differential operator and invoking the identity given in Eq. (6), the integral constitutive relation is transformed into the following partial differential equation that governs the evolution of the nonlocal stress field σ_{ij}

$$L\sigma_{ij} = \sigma_{ij}^L, \quad (7)$$

where the classical Cauchy stress σ_{ij}^L appears as a forcing (nonhomogeneous) term in the field equation. This formulation confirms that the attenuation kernel α plays the role of the Green's function corresponding to the selected operator L , thereby converting the original integral constitutive law into an equivalent differential

representation. The operator L is further taken to be linear with constant coefficients. By using Eq. (1) with Eq. (7), the governing equation of motion

$$L(\rho \partial_t^2 u_i) - \partial_j \sigma_{ij}^L = L F_i. \quad (8)$$

In the present framework, the classical Helmholtz operator is extended to a dynamic setting and modeled as a second-order hyperbolic linear differential operator, as⁴⁶

$$L \equiv 1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}. \quad (9)$$

Here, ∇^2 denotes the Laplacian, while ξ is the intrinsic material length parameter governing spatial nonlocal interactions. The effective internal length scale is defined through $\xi = e_0 a_{cl}$, where e_0 is a material constant and a represents the lattice spacing. In an analogous manner, $\tau \in \mathbb{R}^+$ characterizes the characteristic time associated with temporal memory effects. Defining a nonlocal wave speed as $c \in \mathbb{R}^+$ as $c = \xi/\tau$, the resulting operator L assumes the structure of a Klein–Gordon–type operator. As suggested by Eringen⁴⁸, the estimate $\xi \simeq 0.39a$ provides a physically meaningful measure of the spatial interaction range.

- **Constitutive relations**^{49,50}

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \sigma_{ij} = \sigma_{ij}^L = c_{ijkl} e_{kl} + \gamma_{ij} \psi - \eta_{ijk} E_k - \beta_{ij}^T \theta, \quad (10)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \sigma_i^1 = (\sigma_i^1)^L = h_{ij} E_j + A_{ij} \psi_{,j}, \quad (11)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \zeta = \zeta^L = -\gamma_{ij} e_{ij} - g_k E_k - a_1 \psi + \nu \theta, \quad (12)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) D_i = D_i^L = \eta_{ijk} e_{jk} - g_i \psi - h_{ij} \psi_{,j} + \varepsilon_{ij} E_j + \tau_i \theta, \quad (13)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \rho S^* = (\rho S^*)^L = \beta_{ij}^T e_{ij} + \tau_i E_i + \nu \psi + \frac{\rho C_E \theta}{\theta_0}. \quad (14)$$

- **Equations of motion**

$$\sigma_{ij,j}^L + \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) F_i = \rho \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \ddot{u}_i, \quad (15)$$

$$D_{i,i} = 0, \quad \text{where} \quad E_i = -\varphi_{,i}, \quad (16)$$

$$(\sigma_{i,i}^1)^L + \rho \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) l_1^* + \zeta^L = \chi \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \ddot{\psi}. \quad (17)$$

- **Energy scale equation**

$$\rho \theta_0 \dot{S}^* = -q_{i,i}^*. \quad (18)$$

- **Nonlocal higher order fractional TPL heat transfer law**⁵¹

$$\begin{aligned} & \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \left(1 + \sum_{m=1}^n \frac{\tau_q^{m\alpha}}{\Gamma(m\alpha + 1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}}\right) \dot{q}_i^* \\ &= -K_{ij} \frac{\partial}{\partial t} \left(1 + \sum_{m=1}^p \frac{\tau_T^{m\alpha}}{\Gamma(m\alpha + 1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}}\right) \theta_{,i} \\ & \quad - K_{ij}^* \left(1 + \sum_{m=1}^l \frac{\tau_v^{m\alpha}}{\Gamma(m\alpha + 1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}}\right) \theta_{,i} \end{aligned} \quad (19)$$

- **Nonlocal higher order fractional TPL heat model**

From Eqs. (14), (18), and (19), it follows that

$$\begin{aligned}
& K_{ij} \frac{\partial}{\partial t} \left(1 + \sum_{m=1}^p \frac{\tau_T^{m\alpha}}{\Gamma(m\alpha+1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}} \right) \theta_{,ij} + K_{ij}^* \left(1 + \sum_{m=1}^l \frac{\tau_v^{m\alpha}}{\Gamma(m\alpha+1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}} \right) \theta_{,ij} \\
& = \left(1 + \sum_{m=1}^n \frac{\tau_q^{m\alpha}}{\Gamma(m\alpha+1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}} \right) (\rho C_E \ddot{\theta} + \theta_0 (\beta_{ij} \ddot{u}_{i,j} - \tau_i \ddot{\varphi}_{,i} + \nu \ddot{\psi}))
\end{aligned} \quad (20)$$

where $i, j, k, l = 1 - 3$, and the higher orders $p, l, n \in \mathbb{N}$. The σ_{ij}^L , $(\sigma_i^1)^L$, ζ^L , D_i^L , and $(\rho S^*)^L$ are linked with local thermal piezoelectric medium with voids, $\Gamma(\cdot)$ is the gamma function, and α is the fractional order parameter.

Limiting case

The higher-order fractional TPL model can be converted into another form of heat conduction model under particular limiting conditions when:

- If $\tau_q = \tau_v = \tau_T = 0$, and $K_{ij}^* = 0$ then corresponds to Biot⁵² model.
- If $n = 1$, $\tau_q > 0$, $\tau_T = \tau_v = 0$, $K_{ij}^* = 0$, and $\alpha = 1$ then LS model⁵³.
- If $n = 2$, $p = 1$, $K_{ij}^* = \tau_v = 0$, and $\alpha = 1$ then DPL model⁵⁴.
- If $n = 2$, $p = l = 1$, and $\alpha = 1$ corresponds to TPL model⁵⁵.

Model assumptions and modifications

The present theoretical formulation is developed under the following assumptions:

- The material medium is considered a porous piezo-thermo-electric semi-infinite half-space, bounded by a traction-free surface.
- The medium is assumed homogeneous in composition, while porosity effects are incorporated through void volume fraction parameters.
- Spatio-temporal (KG) nonlocality is incorporated to account for microstructural length-scale effects and time-dependent material interactions.
- The porosity behavior of the medium is modeled within the Cowin–Nunziato theoretical framework, accounting for void distribution and equilibrated stress fields.
- Heat conduction is governed by a fractional higher-order three-phase-lag model to represent thermal relaxation and memory-dependent heat transport.
- Surface wave propagation is analyzed under traction-free mechanical boundary conditions, along with electrically open and thermally specified surface constraints.
- Body forces, internal heat sources, and external electric excitations are neglected to focus on intrinsic wave characteristics.
- All field variables are assumed to vary harmonically along the surface direction and decay exponentially with depth into the medium.

Formulation of the problem

We consider the Rayleigh-type surface waves in a semi-infinite solid described by a KG nonlocal piezo-thermo-electric half-space with voids. The material response is governed by a higher-order fractional TPL heat conduction law together with a Klein–Gordon type nonlocal elasticity kernel, which extends beyond the classical based formulations. The geometry of the model is formulated in a Cartesian coordinate framework. The boundary of the half-space corresponds to $z = 0$, while the x -axis is aligned with the intended path of wave motion. The positive z -direction extends vertically into the medium. Figure 1 presents a schematic view of the problem domain. For surface-wave motion restricted to the xz -plane, the relevant field quantities consist of the displacement vector $\vec{u} = (u(x, z, t), 0, w(x, z, t))$, the electric potential $\varphi = \varphi(x, z, t)$, the void volume fraction $\psi = \psi(x, z, t)$, and the temperature field $\theta = \theta(x, z, t)$. These fields evolve according to the governing equations derived under the adopted nonlocal fractional piezo-thermo-poroelastic theory.

Accordingly, the stress–strain relations together with the electric displacement laws, originally expressed in Eqs. (10)–(13), are reformulated in the present KG nonlocal setting as

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2} \right) \sigma_{xx} = \sigma_{xx}^L = c_{11} u_{,x} + c_{13} w_{,z} + \gamma_1 \psi + \eta_{31} \varphi_{,z} - \beta_1^T \theta, \quad (21)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2} \right) \sigma_{xz} = \sigma_{xz}^L = \eta_{15} \varphi_{,x} + c_{55} (u_{,z} + w_{,x}), \quad (22)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2} \right) \sigma_{zz} = \sigma_{zz}^L = c_{13} u_{,x} + c_{33} w_{,z} + \gamma_3 \psi + \eta_{33} \varphi_{,z} - \beta_3^T \theta, \quad (23)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2} \right) \sigma_x^1 = (\sigma_x^1)^L = -h_1 \varphi_{,x} + A_1 \psi_{,x}, \quad (24)$$

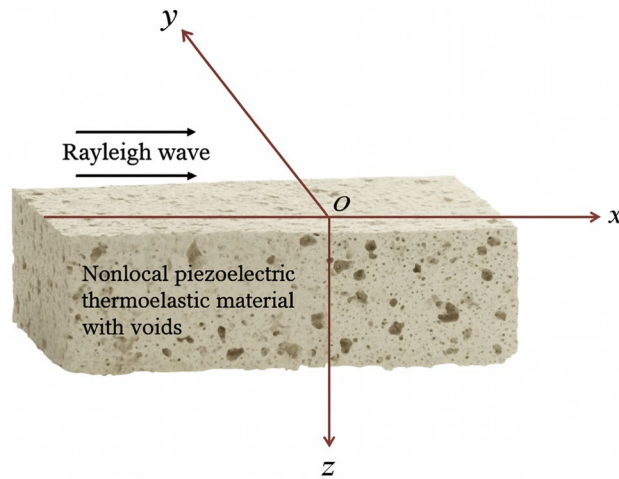


Fig. 1. Schematic layout of the problem.

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \sigma_z^1 = (\sigma_z^1)^L = -h_3 \varphi_{,z} + A_3 \psi_{,z} \quad (25)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) D_x = D_x^L = \eta_{15} (u_{,z} + w_{,x}) - h_1 \psi_{,x} - \varepsilon_{11} \varphi_{,x}, \quad (26)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) D_z = D_z^L = \eta_{31} u_{,x} + \eta_{33} w_{,z} - g_3 \psi - h_3 \psi_{,z} - \varepsilon_{33} \varphi_{,z} + \tau_3 \theta, \quad (27)$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}$.

Incorporating Eqs. (21) to (27), the motion Eqs. (15) to (17), ignoring body forces, can be expressed as

$$c_{11} \frac{\partial^2 u}{\partial x^2} + c_{55} \frac{\partial^2 u}{\partial z^2} + (c_{13} + c_{55}) \frac{\partial^2 w}{\partial x \partial z} + \gamma_1 \frac{\partial \psi}{\partial x} - \beta_1^T \frac{\partial \theta}{\partial x} + (\eta_{31} + \eta_{15}) \frac{\partial^2 \varphi}{\partial x \partial z} = \rho \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \frac{\partial^2 u}{\partial t^2}, \quad (28)$$

$$(c_{13} + c_{55}) \frac{\partial^2 u}{\partial x \partial z} + c_{55} \frac{\partial^2 w}{\partial x^2} + c_{33} \frac{\partial^2 w}{\partial z^2} + \gamma_3 \frac{\partial \psi}{\partial z} - \beta_3^T \frac{\partial \theta}{\partial z} + \eta_{15} \frac{\partial^2 \varphi}{\partial x^2} + \eta_{33} \frac{\partial^2 \varphi}{\partial z^2} = \rho \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \frac{\partial^2 w}{\partial t^2}, \quad (29)$$

$$\eta_{15} \frac{\partial^2 w}{\partial x^2} + \eta_{33} \frac{\partial^2 w}{\partial z^2} + (\eta_{31} + \eta_{15}) \frac{\partial^2 u}{\partial x \partial z} - g_3 \frac{\partial \psi}{\partial z} - h_1 \frac{\partial^2 \psi}{\partial x^2} - h_3 \frac{\partial^2 \psi}{\partial z^2} - \varepsilon_{11} \frac{\partial^2 \varphi}{\partial x^2} - \varepsilon_{33} \frac{\partial^2 \varphi}{\partial z^2} + \tau_3 \frac{\partial \theta}{\partial z} = 0, \quad (30)$$

$$A_1 \psi_{,xx} + A_3 \psi_{,zz} - h_1 \varphi_{,xx} - h_3 \varphi_{,zz} - g_1 \varphi_{,x} - g_3 \varphi_{,z} - \gamma_1 u_{,x} - \gamma_3 w_{,z} - a_1 \psi + \nu \theta = \chi \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \ddot{\psi}. \quad (31)$$

Equation (20) leads to the reduced form of the higher-order TPL fractional heat equation in xz -plane as

$$\begin{aligned} & \left(1 + \sum_{m=1}^p \frac{\tau_T^{m\alpha}}{\Gamma(m\alpha + 1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}}\right) (K_1 \dot{\theta}_{,xx} + K_3 \dot{\theta}_{,zz}) + \left(1 + \sum_{m=1}^l \frac{\tau_v^{m\alpha}}{\Gamma(m\alpha + 1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}}\right) (K_1^* \theta_{,xx} + K_3^* \theta_{,zz}) \\ & = \left(1 + \sum_{m=1}^n \frac{\tau_q^{m\alpha}}{\Gamma(m\alpha + 1)} \frac{\partial^{m\alpha}}{\partial t^{m\alpha}}\right) (\rho C_E \ddot{\theta} + \theta_0 (\beta_1 \ddot{u}_{,x} + \beta_3 \ddot{u}_{,z} - \tau_3 \ddot{\varphi}_{,z} + \nu \ddot{\psi})), \end{aligned} \quad (32)$$

where, $\beta_{ij}^T = \beta_i^T \delta_{ij}$, $K_{ij}^* = K_i^* \delta_{ij}$, $\gamma_{ij} = \gamma_i \delta_{ij}$, $h_{ij} = h_i \delta_{ij}$, $K_{ij} = K_i \delta_{ij}$, $A_{ij} = A_i \delta_{ij}$.

Solution methodology

Assuming wave propagation to occur in the x -direction and to vary with time t , the displacement, thermal, void, and electric potential fields take the form

$$(u, w, \psi, \theta, \varphi) = (\tilde{u}, \tilde{w}, \tilde{\psi}, \tilde{\theta}, \tilde{\varphi}) \exp[\iota \omega (qx + sz - t)] \quad (33)$$

where q and s are the slowness parameters the tilde quantities denote field amplitudes, and ω is the angular frequency.

When Eq. (33) is substituted into Eqs. (28) to (32), the following set of equations is obtained

$$(m_{11} + m_{12}s^2)\tilde{u} + (m_{13}s)\tilde{w} - (m_{14})\tilde{\theta} + (m_{15}s)\tilde{\varphi} + (m_{16})\tilde{\psi} = 0, \quad (34)$$

$$(m_{21}s)\tilde{u} + (m_{22} + m_{23}s^2)\tilde{w} - (m_{24}s)\tilde{\theta} + (m_{25} + m_{26}s^2)\tilde{\varphi} + (m_{27}s)\tilde{\psi} = 0, \quad (35)$$

$$(m_{31})\tilde{u} + (m_{32}s)\tilde{w} + (m_{33} + m_{34}s^2)\tilde{\theta} - (m_{35}s)\tilde{\varphi} - (m_{36})\tilde{\psi} = 0, \quad (36)$$

$$(m_{41}s)\tilde{u} + (m_{42} + m_{27}s^2)\tilde{w} + (m_{43}s)\tilde{\theta} - (m_{44} + m_{45}s^2)\tilde{\varphi} - (m_{46}s + m_{47}s^2 + m_{48})\tilde{\psi} = 0, \quad (37)$$

$$(m_{51})\tilde{u} + (m_{52})\tilde{w} + (m_{53})\tilde{\theta} + (m_{54} + m_{47}s + m_{55}s^2)\tilde{\varphi} + (m_{56} + m_{57}s^2)\tilde{\psi} = 0, \quad (38)$$

where associated coefficients are written in Appendix.

A combination and reduction of Eqs. (34)–(38) lead to the derivation of the characteristic equation given below

$$\sum_{n=0}^{10} \Upsilon_n s^n = 0. \quad (39)$$

The coefficients Υ_n , ($n = 0$ to 10) are found by applying conventional methods, which involve calculating the determinant of the matrix composed of the coefficients from Eqs. (34)–(38). The resulting ten roots, obtained through the application of the Durand–Kerner algorithm in Eq. (39), appear as complex conjugate pairs. These roots are subsequently arranged in accordance with the radiation condition for a half-space, $z > 0$, ensuring $\Im(s_n) > 0$ where disturbances must decay with increasing depth during propagation. Consequently, the five selected values of s (denoted as s_1, s_2, s_3, s_4 , and s_5) correspond to five slowness vectors (q, s_n) , $n = 1 - 5$, each representing a distinct decaying wave phase propagating through the KG nonlocal medium with voids.

Hence, the formal solution adhering to the radiation condition can be represented as follows:

$$(u, w, \varphi, \psi, \theta) = \sum_{m=1}^5 (1, P_m, S_m, V_m, R_m) \tilde{u}_m \exp[\iota\omega(qx + s_m z - t)], \quad (40)$$

where P_m, R_m, S_m , and V_m can be calculated as follows

$$\begin{bmatrix} P_m \\ R_m \\ S_m \\ V_m \end{bmatrix} = \begin{bmatrix} m_{13}s_m & -m_{14} & m_{15}s_m & m_{16} \\ m_{22} + m_{23}s_m^2 & -m_{24}s_m & m_{25} + m_{26}s_m^2 & m_{27}s_m \\ m_{32}s_m & m_{33} + m_{34}s_m^2 & -b_{35} & m_{36} \\ m_{42} + m_{27}s_m^2 & m_{43}s_m & -m_{44} - m_{45}s_m^2 & -m_{46}s_m - m_{47}s_m^2 - m_{48} \end{bmatrix}^{-1} \begin{bmatrix} -m_{11} - m_{12}s_m^2 \\ -m_{21}s_m \\ -m_{31} \\ -m_{41}s_m \end{bmatrix} \quad (41)$$

By means of Eq. (40), Eqs. (23), (24), (26), and (28) are transformed as follows

$$(\sigma_{xz}, \sigma_{zz}, \sigma_z^1, D_z) = \sum_{r=1}^5 \iota\omega (\Pi_{1r}, \Pi_{2r}, \Pi_{3r}, \Pi_{4r}) \tilde{u}_r \exp[\iota\omega(qx + s_r z - t)], \quad (42)$$

where

$$\begin{aligned} \Pi_{1r} &= \Pi_{1r}^L [1 - \tau^2 \omega^2 - \xi^2 \omega^2 (q^2 + s_r^2)], \\ \Pi_{2r} &= \Pi_{2r}^L [1 - \tau^2 \omega^2 - \xi^2 \omega^2 (q^2 + s_r^2)], \\ \Pi_{3r} &= \Pi_{3r}^L [1 - \tau^2 \omega^2 - \xi^2 \omega^2 (q^2 + s_r^2)], \\ \Pi_{4r} &= \Pi_{4r}^L [1 - \tau^2 \omega^2 - \xi^2 \omega^2 (q^2 + s_r^2)], \\ \Pi_{1r}^L &= c_{55} (s_r + qP_r) + \eta_{15} qS_r, \\ \Pi_{2r}^L &= c_{13}q + c_{33}s_r P_r + \eta_{33} S_r s_r - \beta_3^T R_r + \gamma_3 V_r, \\ \Pi_{3r}^L &= A_3 s_r V_r - h_3 s_r S_r \\ \Pi_{4r}^L &= \eta_{31}q + \eta_{33}s_r P_r - \varepsilon_{33}s_r S_r + \tau_3 R_r - g_3 V_r - h_3 s_r V_r. \end{aligned}$$

Boundary conditions

At free surface ($z = 0$) of the thermal piezoelectric medium with KG nonlocality subjected to boundary conditions described as follows

- Mechanical conditions:

$$\sigma_{xz} = 0, \quad \sigma_{zz} = 0, \quad (43)$$

- Equilibrated stresses condition:

$$\sigma_z^1 = 0, \quad (44)$$

- Dielectrical displacement condition:

$$D_z = 0, \quad (45)$$

- Thermal condition:

$$\theta = 0. \quad (46)$$

Derivation of the secular equation

The Rayleigh-type dispersion relation is derived by imposing boundary conditions on the linearized motion equations of the material. A coupled system of simultaneous equations is derived by combining Eqs. (40) and (42) with Eqs. (43) to (46)

$$\sum_{r=1}^5 \Pi_{jr} \tilde{u}_r = 0, \quad (j = 1 - 5), \quad (47)$$

where $\Pi_{5r} = R_r$.

For a meaningful (non-zero) solution to exist, the coefficient matrix $\Pi_{5 \times 5}$ must be singular, i.e., its determinant must vanish. This requirement naturally yields the complex secular equation characteristic of Rayleigh-type surface wave motion. The complex root, denoted by c , is obtained through the Newton–Raphson iterative method. The corresponding phase velocity extracted from c is frequency dependent, demonstrating that Rayleigh-type waves in a KG nonlocal thermo-piezoelectric porous medium are intrinsically dispersive. The presence of a complex c implies attenuation, with its positive imaginary component signifying exponential amplitude decay with depth and energy dissipation within the material. Consequently, the phase velocity and attenuation coefficient of Rayleigh type surface waves are determined using the following expressions

- Phase velocity:

$$V = \frac{|c^2|}{\Re(c)}. \quad (48)$$

- Attenuation coefficient:

$$Q^{-1} = \frac{\Im(1/c^2)}{\Re(1/c^2)}. \quad (49)$$

Trajectory of surface particles

To study the particle motion inside the considered material, the displacement expressions given by Eq. (40) are evaluated for the corresponding value of c . Accordingly, the mechanical displacement components are given as follows

$$(u, w) = \sum_{m=1}^5 (1, P_m) \tilde{u}_m \exp[\imath \omega (qx + s_m z - t)] \quad (50)$$

when q assumes a real value, the corresponding displacements derived from Eqs. (50) can be represented as

$$(U, W) = \Re(u, w) \approx \Re \left(\sum_{m=1}^5 (1, P_m) \exp[\imath \omega (qx + s_m z - t)] \right). \quad (51)$$

The parametric form given in Eq. (51) describes a closed trajectory that characterizes the motion of particles as a function of both time and position. This motion results from the propagation of a surface wave in the piezoelectric medium. For a specified value of $q = \cos(\varphi)/c(\varphi)$, Eq. (51) illustrates the dependence of particle displacement (U, W) on the propagation direction. The scalar term $\Lambda = \sqrt{U^2 + W^2}$, which regulates the magnitude of particle trajectory (*for*), and is inherently linked to the surface wave behavior.

Special cases

- If the influence of fractional order, KG nonlocality, and voids are omitted, the proposed model simplifies to the formulation reported by Sharma et al.⁵⁶.
- Further neglecting the impact of temperature, voids, fractional effects, and KG nonlocal interactions reduces the model to the classical case analyzed by Sharma⁵⁷.

Solution algorithm

The earlier derivations describe the coupled influence of KG nonlocality, thermoelastic parameters, void, and piezoelectric coupling, forming the analytical basis for Rayleigh type wave analysis. The developed algorithm provides a systematic procedure to determine the phase velocity, attenuation coefficient, and surface particle trajectories in the considered medium.

- Beginning with the specified horizontal slowness q , the analysis involves solving a tenth-degree Eq. (39). The Durand–Kerner iterative technique is then used to compute all ten complex roots of s . From this set, five roots corresponding to the decaying phase $\Im(s) > 0$, are selected for further analysis.
- To determine the amplitude ratios P_m, Q_m, R_m , and S_m the equations presented in (41) solved for slowness vectors (q, s_m) .
- Assume a Rayleigh type surface wave propagating at an angle ϕ with the boundary of considered material. Define $q = \cos \phi / c$, where c represent the undetermined wave velocity.
- A determinantal equation in c is obtained by employing P_m, Q_m, R_m, S_m together with the slowness parameter $q = \cos \phi / c$ in the boundary condition equations.
- Using P_m, Q_m, R_m, S_m together with the slowness parameter $q = \cos \phi / c$ in the Eqs. (43) to (46) yields a determinantal equation in c .
- Subsequently, the Newton–Raphson iterative method is applied to compute the wave velocity c . The corresponding attenuation and phase velocity of the Rayleigh-type surface wave are then evaluated using Eqs. (48) and (49).
- Based on Eq. (51), MATLAB is applied to visualize the temporal trajectories of surface particles corresponding to displacement components. The motion is depicted through 2D and 3D graphical representations for enhanced clarity.

Numerical investigation and discussion

The computational scheme outlined earlier serves as a systematic tool for analyzing the characteristics of Rayleigh-type surface waves in the considered medium. Although the derived mathematical expressions are highly coupled due to the simultaneous incorporation of porosity, piezoelectricity, spatio-temporal nonlocality, and fractional higher-order thermal effects, the primary objective of the numerical study is to provide a clear qualitative interpretation of the underlying wave propagation behavior. In the numerical implementation, Cadmium Selenide is adopted as the representative material medium, and the corresponding material constants are taken from previously published studies^{49,58} and listed in Table 1.

The governing equations employed in the present work are formulated in dimensional form. The spatial nonlocal parameter ξ represents an effective characteristic interaction length associated with the porous microstructure and internal nonlocal interactions of the medium, whereas the temporal nonlocal parameter τ and the thermal phase-lag quantities characterize the corresponding thermal memory time scales. The adopted numerical values are selected primarily to examine the qualitative influence of spatio-temporal nonlocality, higher-order thermal memory, and fractional effects on the propagation characteristics of Rayleigh-type surface waves. Therefore, the selected parameter combinations should be interpreted as generalized phenomenological parameters commonly used in nonlocal Sthermoelastic wave propagation analyses rather than as a direct representation of a single experimental configuration.

Symbol	Value	Symbol	Value
c_{11}	$74.1 \times 10^9 \text{ Nm}^{-2}$	c_{33}	$83.6 \times 10^9 \text{ Nm}^{-2}$
c_{13}	$39.3 \times 10^9 \text{ Nm}^{-2}$	T_0	298 K
A_1	$1.3 \times 10^{-5} \text{ N}$	h_1	$4 \times 10^{-16} \text{ Cm}^{-2}$
A_3	$1.2 \times 10^{-5} \text{ N}$	h_3	$5 \times 10^{-16} \text{ Cm}^{-2}$
γ_1	$0.9 \times 10^{10} \text{ Nm}^{-2}$	g_1	$4 \times 10^{-10} \text{ Cm}^{-2}$
γ_3	$0.7 \times 10^{10} \text{ Nm}^{-2}$	g_3	$6 \times 10^{-10} \text{ Cm}^{-2}$
a_1	$1.2 \times 10^{10} \text{ Nm}^{-2}$	ν	$60 \text{ Nm}^{-2} \text{ K}^{-1}$
c_{55}	$13.2 \times 10^9 \text{ Nm}^{-2}$	K_3	$9 \text{ Wm}^{-1} \text{ K}^{-1}$
β_1^T	$6.21 \times 10^5 \text{ Nm}^{-2} \text{ K}^{-1}$	β_3^T	$5.51 \times 10^5 \text{ Nm}^{-2} \text{ K}^{-1}$
η_{31}	-0.160 Cm^{-2}	η_{33}	0.347 Cm^{-2}
η_{15}	-0.138 Cm^{-2}	ε_{11}	$8.26 \times 10^{-11} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
ε_{33}	$9.03 \times 10^{-11} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$	τ_3	$-2.9 \times 10^{-6} \text{ Cm}^{-2} \text{ K}^{-1}$
K_1	$9 \text{ Wm}^{-1} \text{ K}^{-1}$	χ	$1.45 \times 10^{-16} \text{ Nm}^{-2} \text{ s}^2$
K_1^*	$7 \text{ Wm}^{-1} \text{ K}^{-1} \text{ s}^{-1}$	K_3^*	$8 \text{ Wm}^{-1} \text{ K}^{-1} \text{ s}^{-1}$
ρ	5504 kgm^{-3}	C_E	$260 \text{ Jkg}^{-1} \text{ K}^{-1}$

Table 1. Values of the material parameters.

A key strength of the generalized formulation is that it can incorporate higher order fractional-order heat transfer characterized by the parameters n , p and α , offering greater flexibility owing to the selection of these parameters. Chiriță et al.⁵⁹ demonstrated that $p \geq 5$ or $n \geq 5$ results to an unstable system that cannot accurately represent physical behavior. Zampoli⁶⁰ emphasized that, for a model to maintain thermodynamic consistency, the order of expansion should not exceed four, assuming that suitable phase-lag time parameters are incorporated in the formulation. Accordingly, in this study, we adopt $\alpha = 0.9$, $n = 2$, $p = 1$, $l = 1$, $\phi = 60^\circ$, together with the effective spatial nonlocal interaction parameter $\xi = 0.001$, temporal nonlocal parameter $\tau = 0.005$ s and phase-lag times $\tau_q = 0.03$ s, $\tau_T = 0.02$ s, $\tau_v = 0.01$ s, ensuring compliance with the stability and compatibility criteria established by Quintanilla and Racke⁶¹.

Effect of spatial nonlocality on phase velocity and attenuation coefficient

In the following subsection, the influence of spatial nonlocality on the phase velocity and attenuation coefficient is examined under the influence of higher order TPL model. To present the results effectively, graphical illustrations are provided in Fig. 2a and b, which depict the characteristic variation of Rayleigh type wave under different values of the spatial nonlocality parameter ($\xi = 0, 0.001, 0.002, 0.003$) in a piezo-thermo-electric medium containing voids. It is important to note that the case $\xi = 0$ corresponds to the absence of spatial nonlocal interactions, whereas the fully local limiting formulation is recovered when both the spatial and temporal nonlocal parameters satisfy $\xi = 0$ and $\tau = 0$.

Figure 2a depicts the profile of phase velocity of Rayleigh waves as a function of angular frequency ω for distinct values of the spatial nonlocal parameter ξ . For all parameter sets considered, the phase velocity shows a nearly linear dependence on the frequency, but the rate of increase becomes more pronounced as ξ rises. This indicates that nonlocal interactions, as characterized by the Klein-Gordon model, promote higher wave speeds for a given frequency. The nonlocal parameter ξ captures the influence of microstructural interactions and spatial memory in the medium, enabling energy to be transmitted across larger distances and thereby accelerating wave propagation. When $\xi = 0$, the medium behaves in a spatial local manner and the phase velocity remains lowest, showing only modest growth with frequency. As ξ is increased to 0.001, 0.002, and then 0.003, each corresponding curve shifts upward, demonstrating that stronger nonlocal effects lead to a pronounced enhancement in Rayleigh wave velocity for the fractional-order three-phase lag model.

Figure 2b displays the corresponding attenuation profiles as frequency increases for the same values of ξ . Attenuation remains close to zero at low frequencies, especially for the $\xi = 0$. However, as ξ grows, attenuation increases rapidly at higher frequencies, with the steepest rise seen for the largest ξ . This behavior reflects the growing energy dissipation due to nonlocality and spatial memory effects at elevated frequencies. A higher nonlocal parameter means the material accommodates broader-range stress and heat interactions, leading to more complex internal damping mechanisms. Consequently, the wave loses energy more quickly as it travels, which becomes most evident at higher frequencies. Together, these results demonstrate that while enhanced nonlocality boosts phase velocity, it also intensifies energy loss through attenuation, highlighting the fundamental effect of the nonlocal parameter.

Effect of temporal nonlocality on phase velocity and attenuation coefficient

In the following subsection, the effect of temporal nonlocality on the wave dynamics. To elucidate the results, graphical illustrations are presented in Figs. 3a and b, which demonstrate the characteristic variations of these parameters for different values of the temporal nonlocality parameter ($\tau = 0, 0.010, 0.015, 0.020$) in a

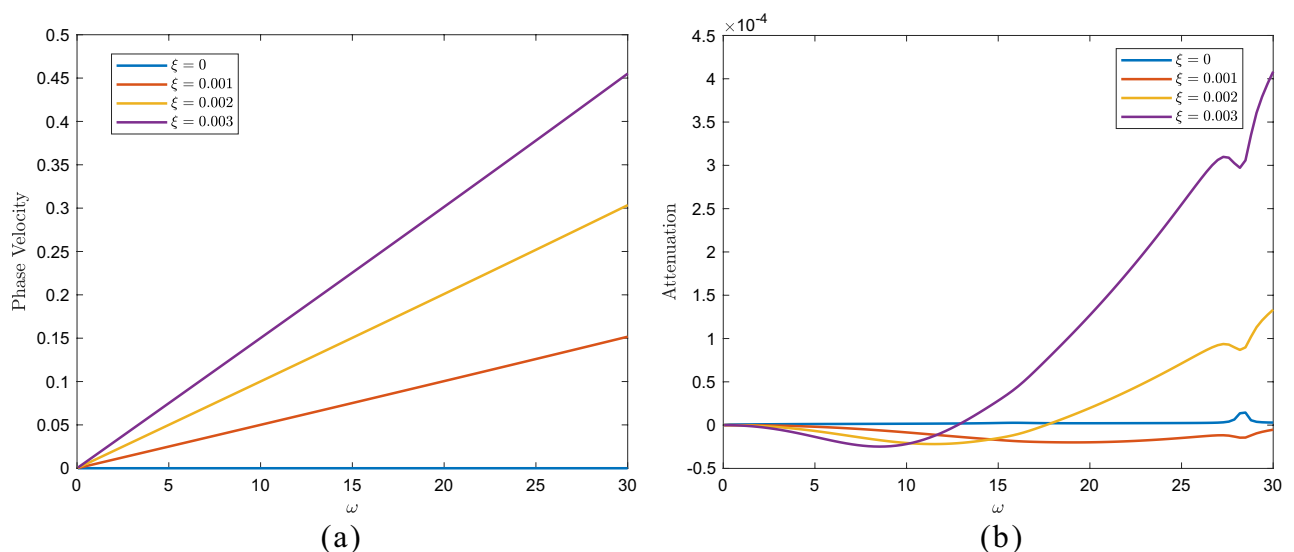


Fig. 2. (a, b) The effect of spatial nonlocality on phase velocity and attenuation with angular frequency (ω).

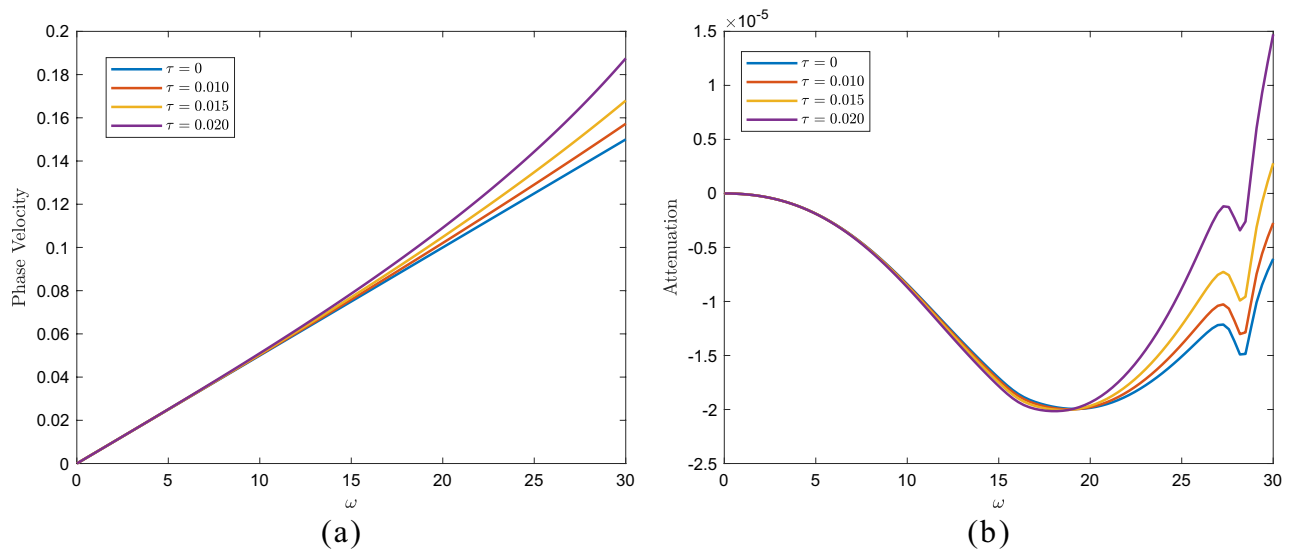


Fig. 3. (a, b) The effect of temporal nonlocality on phase velocity and attenuation with angular frequency (ω).

thermal piezoelectric material with voids. It is noteworthy that the case $\tau = 0$ corresponds to the Eringen-type local formulation, representing the condition where temporal nonlocal effects are neglected.

Figure 3a demonstrates how the temporal nonlocal parameter τ alters the phase velocity of Rayleigh waves. The curves depict that phase velocity increases not only with frequency but also with higher values of τ . When τ is zero, the medium responds locally in time, yielding the lowest wave velocities. As τ increases, the phase velocity grows more rapidly, indicating that nonlocal temporal interactions facilitate a faster mechanical and thermal disturbance propagation. Physically, τ characterizes the memory or inertia imparted by the system's history in the evolution process. A larger τ implies that the present behavior depends more strongly on past states, enabling disturbances to persist and propagate more efficiently. This effect is especially evident at higher frequencies, where the presence of temporal nonlocality amplifies the phase velocity significantly.

Figure 3b shows the frequency dependence of attenuation for various τ values. For low frequencies, attenuation remains close to zero for all τ , but as frequency increases, the curves exhibit a distinct negative dip followed by a sharp rise. The magnitude and sharpness of these features grow with increasing τ . The initial negative attenuation suggests regions where temporal nonlocality may transiently reinforce the amplitude of waves, possibly due to constructive interaction between the present and the memory-influenced past states. As frequency rises further, the enhanced τ drives stronger attenuation, indicating more pronounced dissipation due to nonlocal temporal effects. This behavior highlights how the KG operator's temporal nonlocality governs both the persistence and decay of waves, fundamentally shaping the energy distribution and propagation characteristics in the thermal piezoelectric material described by the fractional three-phase lag model.

Effect of higher order parameter on phase velocity and attenuation coefficient

In the following subsection, the effect of higher-order heat conduction model parameters on the wave dynamics. To elucidate the results, graphical illustrations are presented in Fig. 4a and b, which demonstrate the characteristic variations of these parameters for different values of the higher-order parameters ($n = 4, p = 3, l = 3$), ($n = 3, p = 2, l = 2$), and ($n = 2, p = 1, l = 1$) in a thermo-piezoelectric medium with voids.

Figure 4a shows the behavior of phase velocity as a function of angular frequency ω for varying choices of the higher-order heat conduction model parameters n, p , and l . Each curve corresponds to a distinct combination, where higher values represent a greater inclusion of memory and nonlocality components in the fractional-order heat equation. It is seen that the case with $n = 4, p = 3, l = 3$ gives the highest phase velocity across the frequency domain, while the combination $n = 3, p = 2, l = 2$ remains the lowest, with $n = 2, p = 1, l = 1$ showing intermediate behavior. The observed trend reveals that adding extra higher-order terms to the generalized conduction framework substantially accelerates wave propagation in the material. The inset plot emphasizes a narrow frequency window, further highlighting how the subtle effect of increasing n, p , produces discernible changes in phase velocity even at fine scales. These parameters reflect the amplitude and memory depth of various phases of lag (as introduced via the fractional and multi-term nature of the model), demonstrating that richer nonlocal and hereditary structure within the heat conduction law supports more rapid transfer of energy and momentum.

Figure 4b illustrates attenuation profiles for the same set of higher-order heat conduction parameters. At low frequencies (near $\omega = 0$), all models show nearly identical attenuation, with values remaining close to zero. This means that for slowly varying waves, the choice of higher order terms has minimal influence, and the material dissipates very little wave energy regardless of model details. As the frequency increases up to about $\omega = 15$, the attenuation gently rises and the differences between curves remain small, as shown in the zoomed inset around $\omega = 14$. In this middle frequency range, increasing the order of the heat conduction model slightly changes

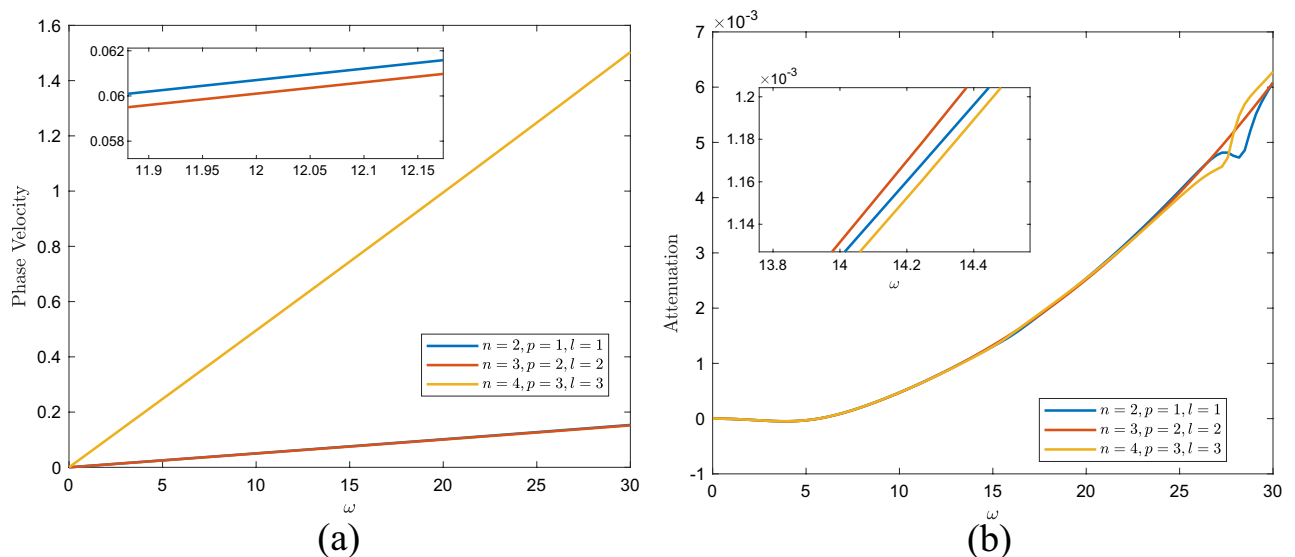


Fig. 4. (a, b) The effect of higher order parameters in heat conduction model on phase velocity and attenuation with angular frequency (ω).

attenuation, but the overall behavior of each model remains close. At higher frequencies ($\omega > 20$), the impact of using a higher order model becomes much more pronounced. The yellow curve, representing the most advanced model ($n = 4, p = 3, l = 3$), which started below the other curves at lower frequencies, now rises rapidly and eventually overtakes the others. The blue and red curves, corresponding to models of lower order, increase less steeply. This crossing pattern shows that the response of the medium is not simply determined by model order alone, but also by frequency: at low and intermediate frequencies, a more advanced model may actually provide less attenuation, effectively supporting the propagation of waves. However, as the frequency gets higher, the same model begins to generate much greater attenuation, causing energy to dissipate more rapidly.

This frequency-dependent behavior occurs because higher-order heat conduction models capture memory and nonlocal effects that become more pronounced at higher thermal and mechanical rates. With more higher order terms included, the medium can respond more sensitively to rapid oscillations, which promotes stronger internal damping at high ω . Thus, the graph clearly demonstrates that while all three models behave similarly at low frequencies, the most advanced higher order model leads to the highest energy dissipation for high-frequency Rayleigh type waves, reflecting an intricate and dynamic connection between model structure and wave attenuation.

Effect of fractional order parameter on phase velocity and attenuation coefficient

In the following subsection, the response of fractional order parameter (α) on the wave dynamics is investigated. To elucidate the results, graphical illustrations are presented in Fig. 5a and b, which demonstrate the effect of distinct values of the fractional order parameter $\alpha = 0.85, \alpha = 0.90, \alpha = 0.95, \alpha = 1.00$ on wave characteristics in a thermo-piezoelectric medium with voids.

Figure 5a depicts the profile of phase velocity of Rayleigh type surface waves in relation to angular frequency ω for distinct values of α . Here, $\alpha = 1$ corresponds to the standard integer-order derivative and $\alpha < 1$ represents fractional derivatives with additional memory and nonlocal effects. It is observed that as α increases from 0.85 toward 1.00, the phase velocity curves exhibit a gradual rise across all frequencies. Lower values of α yield reduced phase velocities, while the standard case $\alpha = 1$ allows the wave to propagate fastest through the medium. This result demonstrates that increasing α enhances the medium's responsiveness to dynamic disturbances, and lower values of α , incorporating stronger memory, slow down the propagation. A higher α means less fractional influence and more direct heat conduction, resulting in faster wave and energy transport, especially at higher frequencies.

Figure 5b represents the trend of the attenuation coefficient of the Rayleigh type surface wave w.r.t. the angular frequency (ω) for distinct values of α . The attenuation exhibits a strongly frequency-dependent behavior, it decreases in magnitude to a pronounced minimum in the mid-frequency band ($\approx 18-20$), shows an anomalous reversal of the α ordering in the upper-mid band ($\approx 24-27$) where smaller α yields slightly larger attenuation, and then returns to the original ordering at the highest frequencies. Thus, the sensitivity of attenuation to the fractional order is not uniform α has negligible effect at very low frequencies, enhances damping for smaller values in the mid-band, produces a transient reversed effect due to dispersive/resonant interactions in the upper-mid band, and finally reverts to the memory-dominated trend at high frequencies. This behaviour indicates a competition between memory-induced dissipation and dispersive/inertial mechanisms that is strongly frequency dependent.

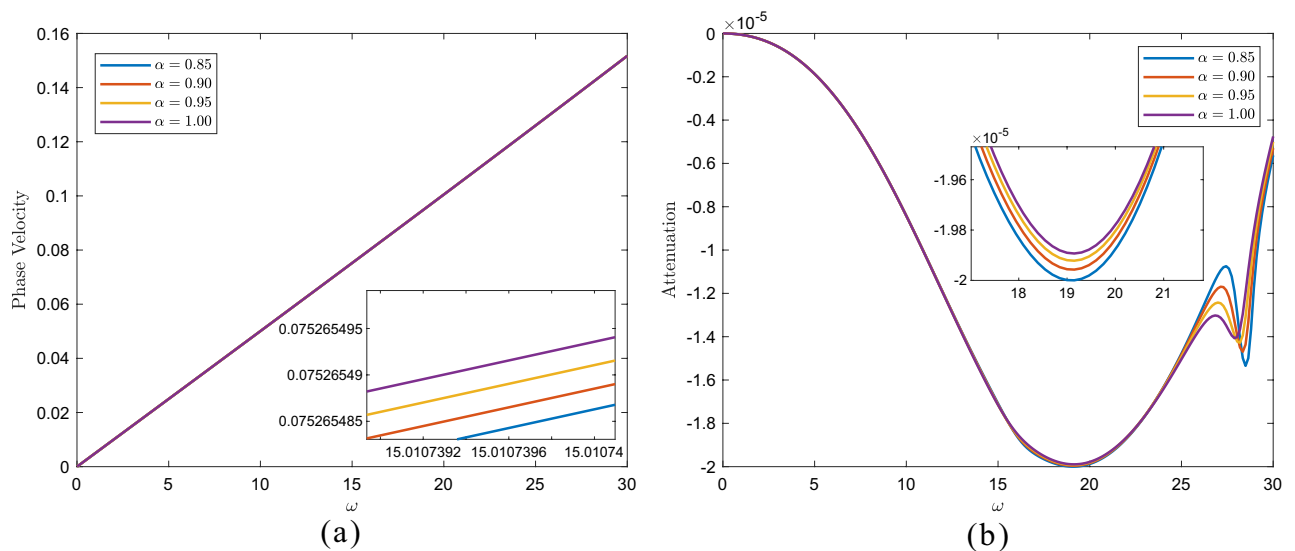


Fig. 5. (a, b) The effect of fractional order parameter on phase velocity and attenuation with angular frequency (ω).

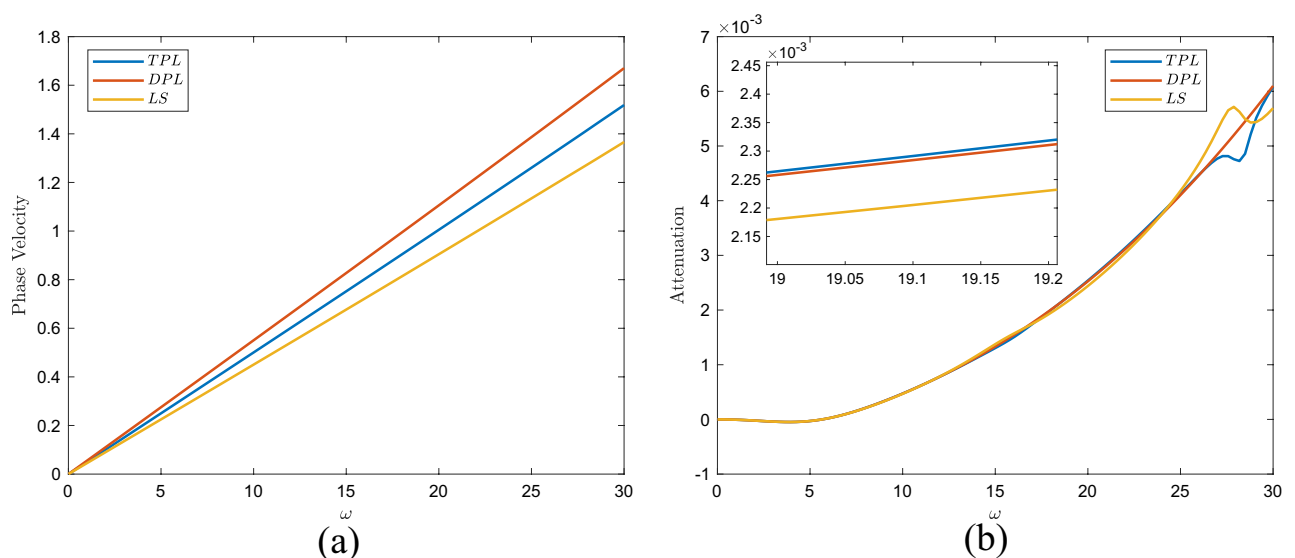


Fig. 6. (a, b) The effect of different heat conduction model on phase velocity and attenuation with angular frequency (ω).

Effect of distinct heat models on phase velocity and attenuation coefficient

In the next subsection, we explore the effects of three heat transfer models on the phase velocity and attenuation coefficient. These outcomes are visualized in Fig. 6a and b, which highlight the wave characteristics under the TPL, DPL, and LS formulations.

In Fig. 6a, the influence of the TPL, DPL, and LS heat-conduction models on the phase velocity of Rayleigh waves is displayed as a function of angular frequency. While each model produces a rising phase-velocity trend with increasing frequency, distinct discrepancies among the models become evident. The DPL model consistently delivers the highest phase velocity across the frequency range, indicating a more responsive material where wave propagation benefits from additional lag features in both heat and thermal fields. The TPL model, though slightly lower than DPL, also shows enhanced velocities compared to the LS model. The LS case, representing the simplest heat conduction theory here, produces the lowest phase velocity, reflecting the more restricted ability of the medium to respond swiftly to fast changes. These distinctions become more noticeable as frequency increases, suggesting that models which include multiple phase lags foster more rapid and efficient energy and wave transport, especially at higher frequencies.

Figure 6b presents the frequency dependence of attenuation for these three heat conduction models. At low frequencies, attenuation is minimal and nearly identical for all three cases, reflecting limited energy loss and underscoring the reduced role of lag effects at slow oscillations. As frequency increases, the attenuation curves begin to separate, with TPL and DPL models demonstrating greater attenuation than the LS model. Examining the attenuation curve closely, it can be seen that the LS model exhibits a characteristic jump or rapid increase in attenuation at higher frequencies compared to the other models. While at low and moderate frequencies, the LS curve remains below the TPL and DPL curves, as frequency approaches its upper limit in the considered range, the LS attenuation sharply rises and even overtakes those of the TPL and DPL models, showing a distinct jump-like behavior. This indicates that for very fast oscillations, the classical LS model becomes significantly more dissipative, resulting in enhanced wave energy loss within the medium. This sudden increase in attenuation for the LS model at high frequency suggests that although it represents a simpler heat conduction law, it manifests pronounced internal damping when the system is driven at very rapid rates. In contrast, TPL and DPL models, which incorporate more advanced nonlocal and lag features, have more tempered or smoother growth in attenuation, reflecting their ability to distribute dissipation mechanisms through additional relaxation channels. The LS model's abrupt rise may be attributed to the absence of higher-order or multiple-phase relaxation effects, causing all dissipation to concentrate at higher frequencies rather than being spread across intermediate regimes.

Overall, the behavior seen in the attenuation curves highlights the importance of selecting an appropriate heat conduction model for accurately predicting damping, especially in high-frequency applications where simpler models like LS might underestimate or suddenly overestimate energy losses.

Influence of depth on particle motion

In this section, six figures, labeled as Fig. 7a–f, provide a detailed visualization of Rayleigh type wave particle motion at various shallow depths under the nonlocal fractional order TPL heat conduction model. The plots systematically capture both the two-dimensional and three-dimensional trajectories traced by particles as the wave propagates through different subsurface levels. For systematic examination, four shallow depth values are selected, and the interval from $1e-05$ to $1.1e-05$ is divided into three equal increments. This fine partitioning allows for highly detailed scrutiny of how subtle changes in depth impact the particle motion characteristics.

Figure 7e synthesizes the data from all four depths by plotting them together, offering a comprehensive visual comparison of particle motion trends across the depth range. The sequence of figures displays both horizontal and vertical particle trajectories in 3D, evolving continuously with time. Figure 7f presents the data from the initial four plots in a 2D format, enabling more precise identification of nuanced variations.

Examination of Fig. 7a–d reveals that Rayleigh type wave particles, at each considered depth, trace out spiral elliptical paths combining both horizontal and vertical components consistent with Rayleigh wave behavior known to cause simultaneous movement in both directions. While the general elliptical form persists, there are distinct differences in amplitude for the horizontal and vertical motions between the different depth values. The observed variability indicates that Rayleigh wave amplitude is sensitive to changes in subsurface depth. Figure 7e confirms that trajectory differences grow with increasing depth, underscoring the depth-specific behavior of Rayleigh type wave propagation.

These results emphasize that Rayleigh wave dynamics are fundamentally depth-dependent. The depth-related phenomena shown across Fig. 7a–f highlight the continuous interplay between wave propagation and the characteristics of the subsurface medium. Variations in the spiral elliptical trajectories demonstrate how Rayleigh waves evolve in different geological layers. Such observations have important practical implications for earthquake engineering, as understanding depth-specific Rayleigh wave behavior is crucial for assessing seismic hazard and supporting the structural stability of buildings and communities.

Validation and comparison with previously published results

To establish the reliability and physical consistency of the present formulation, a comparative validation has been carried out with previously published results available in the literature. The comparison is performed at two levels, namely dispersion characteristics and particle motion behavior associated with Rayleigh wave propagation.

Figure 8a and b presents the validation comparison of normalized phase velocity and normalized attenuation with angular frequency. The normalization has been performed by scaling the computed magnitudes with respect to their corresponding maximum values so that all variations remain confined within the 0–1 range. This approach enables a direct comparison of dispersion trends independent of absolute magnitude differences. It is observed from Fig. 8 that the dispersion trends of the two formulations do not coincide exactly. The variation patterns differ in slope, curvature, and attenuation evolution across the frequency spectrum. This deviation is physically justified due to fundamental differences in the constitutive modeling and material coupling. In the previously published work⁶², the medium consisted of a piezoelectric material coupled with a semiconductor phase, where thermoelastic interactions were governed through the Moore–Gibson–Thompson model and Eringen's spatial nonlocal elasticity⁶³.

In contrast, the present formulation considers a piezoelectric medium coupled with void inclusions instead of semiconductors, while retaining thermoelastic effects. The thermal behavior is described using a fractional three-phase-lag model incorporating higher-order thermal memory. Additionally, the nonlocal response follows the KG-type framework, which introduces both spatial and temporal nonlocality. Owing to these differences in thermal lagging, coupling mechanisms, and nonlocal interactions, quantitative deviations in phase velocity and attenuation are expected. Therefore, the non-coincident dispersion trends observed in Fig. 8 remain physically consistent and theoretically justified.

A more fundamental validation is obtained through the comparison of particle motion trajectories, illustrated in Fig. 9a and b. Rayleigh surface waves are well known to propagate with elliptic particle motion in the sagittal

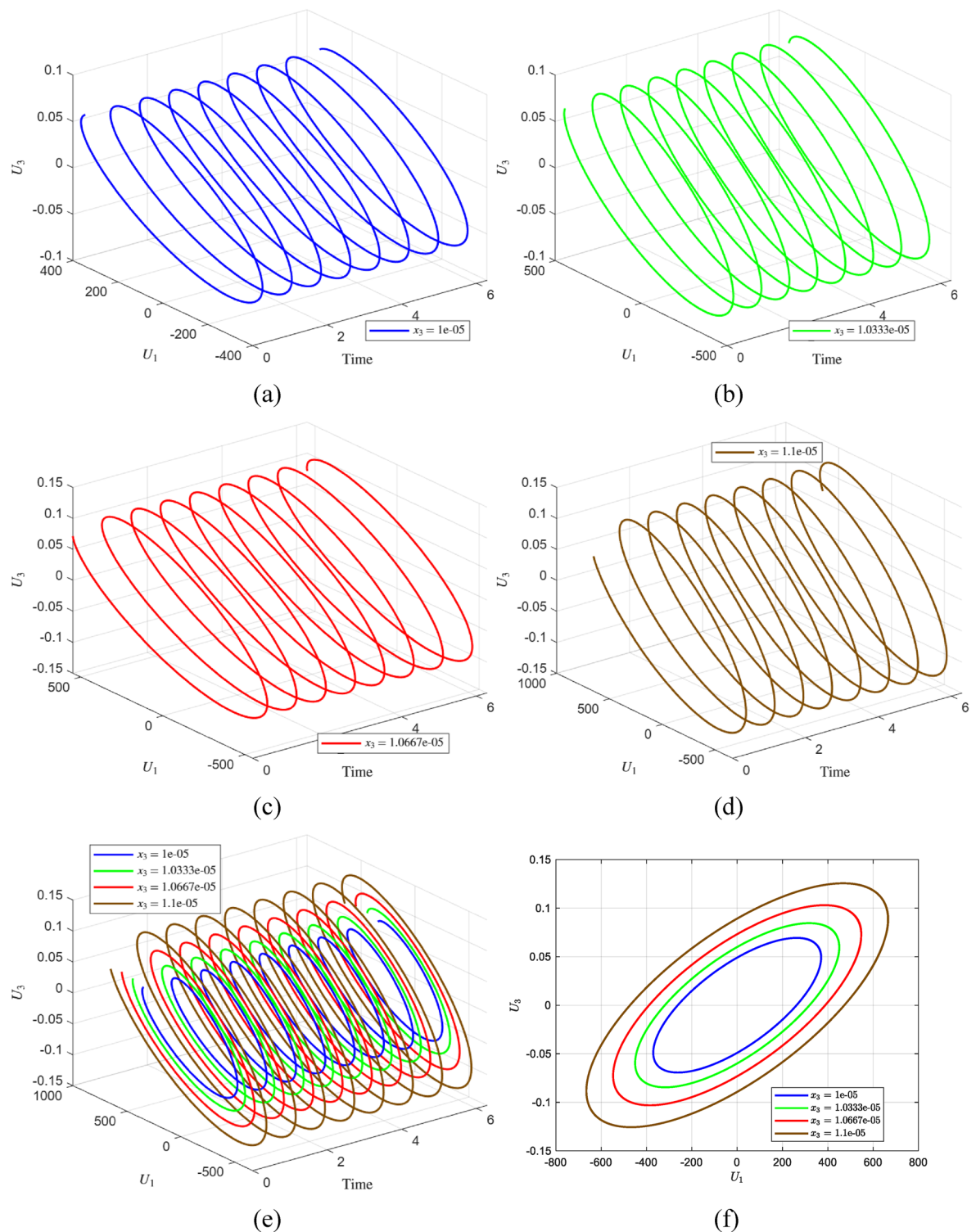


Fig. 7. (a–f) Variation of particle motion (U , W) with time for different shallow depth z values.

plane^{57,64}, where horizontal (U) and vertical (W) displacement components remain phase shifted, producing closed elliptical paths. The particle trajectories plotted for both formulations exhibit clear elliptic motion. Despite differences in coupling phase (semiconductor versus void), thermal modeling, and nonlocal frameworks, the displacement paths align in their fundamental kinematic nature. The trajectories remain confined to the sagittal plane and follow the classical elliptic motion condition of Rayleigh waves without deviation toward circular or longitudinal behavior.

Thus, while Fig. 8 reflects model-dependent dispersion differences, Fig. 9 confirms that the essential propagation physics remains preserved. The inclusion of voids, fractional thermal lagging, and temporal

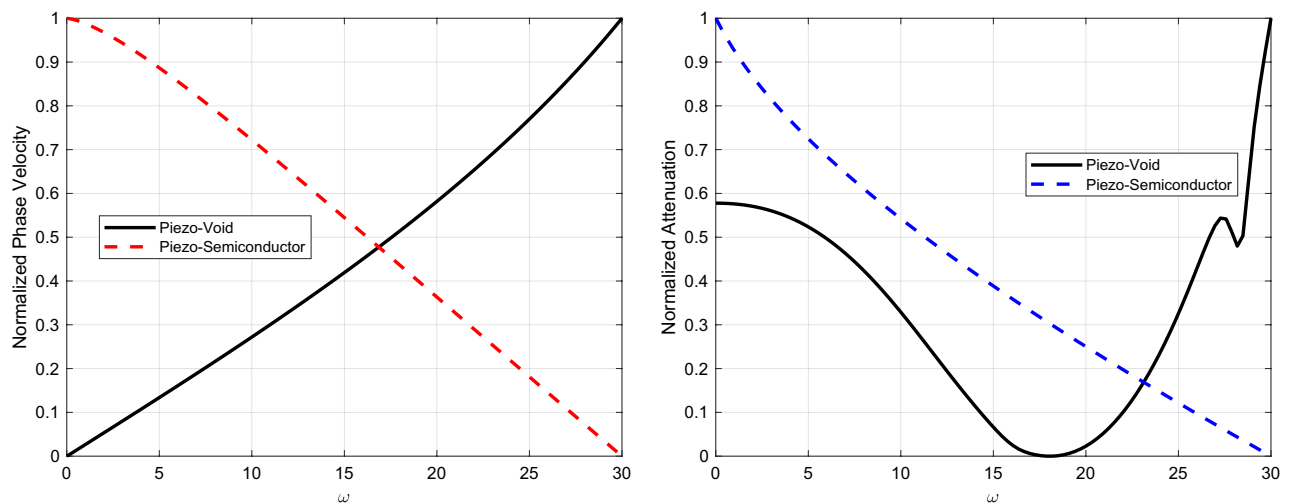


Fig. 8. (a, b) Validation comparison of normalized phase velocity and attenuation versus angular frequency (ω) between the present piezo-void model and the previously published piezo-semiconductor model⁶².

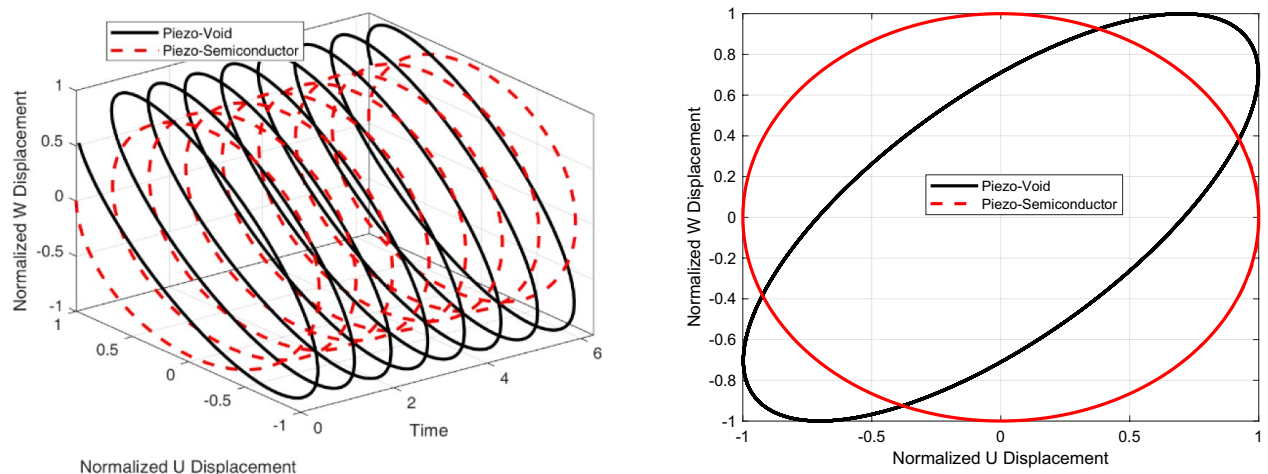


Fig. 9. (a, b) Validation comparison of normalized particle motion trajectories (U–W plane) for Rayleigh wave propagation in the present piezo-void model and the previously published piezo-semiconductor model⁶².

nonlocality influences only the size, inclination, and eccentricity of the particle ellipse rather than altering its intrinsic elliptic character. Overall, the agreement of elliptic particle motion, together with physically explainable dispersion deviations, validates the present formulation and confirms its consistency with established Rayleigh wave propagation characteristics reported in the literature.

Conclusions

This study presents an in-depth analysis of Rayleigh type wave propagation in a generalized thermal piezoelectric semi-infinite medium incorporating voids. The investigation employs a fractional-order Klein–Gordon-type nonlocal TPL framework, wherein the governing field equations are rigorously formulated and solved using an eigenmode analytical approach. The primary objective is to elucidate the coupled influences of spatial and temporal nonlocality, fractional-order effects, heat transfer models, and depth variations on surface wave. A detailed examination of the computational results, as illustrated in Figs. 2, 3, 4, 5, 6, 7, 8 and 9, leads to several significant findings and physical interpretations summarized below.

- The phase velocity of Rayleigh waves increases with the spatial nonlocal parameter, indicating that stronger microstructural coupling under the Klein–Gordon-type nonlocality enhances wave propagation.
- Increasing the spatial nonlocal parameter enhances the overall attenuation at higher frequencies, indicating that spatial nonlocality strengthens internal damping through extended stress–heat coupling and microstructural interactions within the medium.

- The temporal nonlocal parameter τ promotes higher phase velocities and sustained wave propagation, highlighting the role of temporal memory in energy transmission.
- With larger τ , attenuation first dips (transient reinforcement) and then rises sharply at high frequencies, reflecting the dual effects of energy storage and dissipation due to temporal memory.
- Higher fractional heat conduction orders (n , p , l) enhance wave dispersion and energy transport, with the model $n = 4$, $p = 3$, $l = 3$ yielding the maximum phase velocity.
- At high frequencies, the higher-order model exhibits greater attenuation, indicating dominant thermal memory and damping effects.
- The fractional-order parameter α critically governs wave propagation. Lower values of α slightly reduce phase velocity and modify the attenuation behavior across frequencies, while the integer-order case $\alpha = 1$ ensures the fastest wave propagation.
- Among distinct heat conduction models, the DPL model provides the highest phase velocity, followed by TPL and LS; at high frequencies, the LS model exhibits a sharp attenuation rise, reflecting limited multi-relaxation capability.
- Rayleigh wave particle motion shows clear depth dependence, with larger horizontal and vertical displacements at shallower depths. This near-surface amplification reflects localized energy concentration relevant to seismic and geotechnical applications.

The present findings provide useful insights for optimizing surface acoustic wave sensors, seismic characterization systems, and energy-harvesting devices. The model's ability to capture nonlocal and fractional thermal effects strengthens predictions of wave behavior in advanced piezoelectric and porous materials. These results also open up several future research opportunities, particularly in experimental validation, high-fidelity numerical simulations, and broader extensions of the theoretical framework.

Appendix

$$\begin{aligned}
 m_{11} &= c_{11}q^2 - \rho(1 + \omega^2\xi^2q^2 - \tau^2\omega^2), & m_{12} &= c_{55} - \rho\omega^2\xi^2, & m_{13} &= (c_{13} + c_{55})q, \\
 m_{14} &= -\beta_1^Tqb^{-1}, & m_{15} &= (\eta_{31} + \eta_{15})q, & m_{16} &= -\gamma_1qb^{-1}, \\
 m_{21} &= (c_{13} + c_{55})q, & m_{22} &= c_{55}q^2 - \rho(1 + \xi^2\omega^2q^2 - \tau^2\omega^2), & m_{23} &= c_{33} - \rho\xi^2\omega^2, \\
 m_{24} &= -\beta_3^Tb^{-1}, & m_{25} &= \eta_{15}q^2, & m_{26} &= \eta_{33}, & m_{27} &= -\gamma_3b^{-1}, \\
 m_{31} &= \theta_0\beta_1^Tbq, & m_{32} &= \theta_0\beta_3^Tb, & m_{33} &= L_1K_1q^2b + L_2K_1^*q^2 - \rho C_E, \\
 m_{34} &= L_1K_3b + L_2K_3^*, & m_{35} &= \theta_0\tau_3b, & m_{36} &= \theta_0v, & b &= -i\omega, \\
 m_{41} &= (\eta_{31} + \eta_{15})q, & m_{42} &= \eta_{15}q^2, & m_{43} &= -\tau_3b^{-1}, & m_{44} &= \varepsilon_{11}q^2, & m_{45} &= \varepsilon_{33}, \\
 m_{46} &= -g_3b^{-1}, & m_{47} &= h_3, & m_{48} &= h_1q^2, \\
 m_{51} &= \gamma_1qb^{-1}, & m_{52} &= \gamma_3b^{-1}, & m_{53} &= vb^{-2}, & m_{54} &= -h_1q^2 + g_1qb^{-1}, \\
 m_{55} &= g_3b^{-1}, & m_{56} &= q^2A_1 - a_1b^{-2} - \chi(1 + \omega^2\xi^2q^2 - \tau^2\omega^2), & m_{57} &= -\chi\xi^2\omega^2,
 \end{aligned}$$

$$L_1 = \frac{1 + \sum_{m=1}^p \frac{\tau_T^{m\alpha} t^{-m\alpha} E_{1,1-m\alpha}(b)}{\Gamma(m\alpha+1)}}{1 + \sum_{m=1}^n \frac{\tau_q^{m\alpha} t^{-m\alpha} E_{1,1-m\alpha}(b)}{\Gamma(m\alpha+1)}}, \quad L_2 = \frac{1 + \sum_{m=1}^l \frac{\tau_v^{m\alpha} t^{-m\alpha} E_{1,1-m\alpha}(b)}{\Gamma(m\alpha+1)}}{1 + \sum_{m=1}^n \frac{\tau_q^{m\alpha} t^{-m\alpha} E_{1,1-m\alpha}(b)}{\Gamma(m\alpha+1)}},$$

where $E_{\alpha_1, \alpha_2}(z)$ is the Mittag-Leffler function for any $z \in \mathbb{C}$ in two parameters $\alpha_1, \alpha_2 \in \mathbb{C}$ with $\Re(\alpha_1) > 0$ defined as

$$E_{\alpha_1, \alpha_2}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha_1 k + \alpha_2)}.$$

Received: 21 February 2026; Accepted: 19 June 2026

Published online: 16 September 2026

References

1. Bai, X. et al. Omnidirectional sound wave absorption based on the multi-oriented acoustic meta-materials. *Appl. Acoust.* **228**, 110344. <https://doi.org/10.1016/j.apacoust.2024.110344> (2025).
2. Q. Long, G. Jiang, J. Zhou, D. Zhao, and H. Yu, A cellulose ionogel with rubber-like stretchability for low-grade heat harvesting. *Research*. <https://doi.org/10.34133/research.0533> (2024).
3. Phung-Van, P., Tran, L. V., Ferreira, A. J. M., Nguyen-Xuan, H. & Abdel-Wahab, M. Nonlinear transient isogeometric analysis of smart piezoelectric functionally graded material plates based on generalized shear deformation theory under thermo-electro-mechanical loads. *Nonlinear Dyn.* **87**(2), 879–894. <https://doi.org/10.1007/s11071-016-3085-6> (2017).
4. Chao, Y. & Kuo, C.-N. Design criteria and energy absorption prediction empirical rule for deformation-induced densification of gradient porous materials. *Mater. Sci. Addit. Manuf.* **3**(3), 4234. <https://doi.org/10.36922/msam.4234> (2024).
5. Yang, Z. et al. Observer-based fuzzy adaptive dynamic surface force control for pneumatic polishing system end-effector with uncertain contact environment model. *IEEE Trans. Autom. Sci. Eng.* **22**, 17898–17913. <https://doi.org/10.1109/TASE.2025.3585136> (2025).

6. Amini, Y., Sedghamiz, M. A. & Hassanvand, A. Application of halloysite nanotubes (HNTs) in membrane technology for oil separation; A review study. *Mater. Today Commun.* **50**, 114333. <https://doi.org/10.1016/j.mtcomm.2025.114333> (2026).
7. Rayleigh, L. On waves propagated along the plane surface of an elastic solid. *Proc. Lond. Math. Soc.* **s1-17**(1), 4–11. <https://doi.org/10.1112/plms/s1-17.1.4> (1885).
8. Yang, X. et al. Analysis of thermal wave scattering and temperature distribution in sub-surface, defects of gradient construction materials. *Sci. Rep.* **15**(1), 22381 (2025).
9. Wang, X., Li, Z., Qu, Z. & Cheng, R. A high-accuracy ultrasonic flow measurement method based on voltage amplitude difference. *IEEE Trans. Instrum. Meas.* **74**, 1–12. <https://doi.org/10.1109/TIM.2025.3568964> (2025).
10. Zhu, X. et al. Transient modeling and analysis of a stepped-configuration thermoelectric generator considering non-uniform temperature distribution. *Appl. Energy* **383**, 125360. <https://doi.org/10.1016/j.apenergy.2025.125360> (2025).
11. Miao, Z., Meng, X., Li, X. & Liang, B. Influence weights of key parameters and optimization strategies for non-contact thermoelectric generator performance enhancement. *Energy* **339**, 139120. <https://doi.org/10.1016/j.energy.2025.139120> (2025).
12. He, Y. et al. Feasibility study of mechanical stress wave detection in power semiconductor devices using bare FBG sensors. *IEEE Sens. J.* **25**(21), 39849–39857. <https://doi.org/10.1109/JSEN.2025.3615108> (2025).
13. Ma, Z. & Liu, T.-J. Fretting contact of FGPM coating-substrate system under cyclic torque based on laminated model. *J. Intell. Mater. Syst. Struct.* **36**(13), 926–946. <https://doi.org/10.1177/1045389X251352806> (2025).
14. Cai, Y. et al. Nature-inspired wearable thermoelectric generator for body heat harvesting. *Energy* **338**, 138899. <https://doi.org/10.1016/j.energy.2025.138899> (2025).
15. Chen, B. et al. Research on the measurement technology for pretension stress on small-sized bolts based on the piezoelectric ultrasonic resonance method. *Materials* **17**(23), 5802. <https://doi.org/10.3390/ma17235802> (2024).
16. Awad, E., El-Bary, A. A. & Dai, W. Linear damped oscillations underlying the fractional Jeffreys equation. *Fract. Fract.* **9**(9), 556 (2025).
17. Awad, E. Asymptotic analysis for Cauchy stresses in a semi-infinite space with anomalous thermal conduction. *Fract. Calc. Appl. Anal.* <https://doi.org/10.1007/s13540-026-00507-8> (2026).
18. Nowacki, W. Foundations of linear piezoelectricity. *Electromag. Interact. Elast. Solids* **257**, 105–157 (1979).
19. Cowin, S. C. & Nunziato, J. W. Linear elastic materials with voids. *J. Elast.* **13**(2), 125–147. <https://doi.org/10.1007/BF00041230> (1983).
20. Iovane, G. & Nasedkin, A. V. Modal analysis of piezoelectric bodies with voids. I. Mathematical approaches. *Appl. Math. Model.* **34**(1), 60–71. <https://doi.org/10.1016/j.apm.2009.03.028> (2010).
21. Jiang, P., Zheng, H., Xiong, J. & Rabczuk, T. The localized radial basis function collocation method for dendritic solidification, solid phase sintering and wetting phenomenon based on phase field. *J. Comput. Phys.* **520**, 113515. <https://doi.org/10.1016/j.jcp.2024.113515> (2025).
22. Zhang, C., Du, C. & Mohammadzadeh, A. A practical type-3 fuzzy vibration control for high-speed trains subjected to unknown nonlinear dynamics. *J. Frankl. Inst.* **362**(16), 108115. <https://doi.org/10.1016/j.jfranklin.2025.108115> (2025).
23. Zhao, W. et al. TiN supported 3D directional tubular skeleton encapsulating phase change materials for efficient solar-thermal energy conversion and storage. *Sol. Energy* **302**, 114071. <https://doi.org/10.1016/j.solener.2025.114071> (2025).
24. Ullah, Z., Raza, N., Akhtar, H. & Sonil, N. I. Prototype fabrication of graphite-patterned flexible sensors and gauges for liquid level monitoring. *Mater. Chem. Phys.* **348**, 131587. <https://doi.org/10.1016/j.matchemphys.2025.131587> (2026).
25. Edelen, D. G. B. & Laws, N. On the thermodynamics of systems with nonlocality. *Arch. Ration. Mech. Anal.* **43**(1), 24–35. <https://doi.org/10.1007/BF00251543> (1971).
26. Romano, G., Barretta, R., Diaco, M. & Marotti de Sciarra, F. Constitutive boundary conditions and paradoxes in nonlocal elastic nanobeams. *Int. J. Mech. Sci.* **121**, 151–156. <https://doi.org/10.1016/j.ijmecsci.2016.10.036> (2017).
27. Anh, V. & Vinh, P. Expressions of nonlocal quantities and application to Stoneley waves in weakly nonlocal orthotropic elastic half-spaces. *Math. Mech. Solids* **28**(11), 2420–2435. <https://doi.org/10.1177/10812865231164332> (2023).
28. Anh, V. T. N., Vinh, P. C., Tuan, T. T. & Hue, L. T. Weakly nonlocal Rayleigh waves with impedance boundary conditions. *Continuum Mech. Thermodyn.* **35**(5), 2081–2094. <https://doi.org/10.1007/s00161-023-01235-7> (2023).
29. de Oliveira, E. C. & Tenreiro Machado, J. A. A review of definitions for fractional derivatives and integral. *Math. Probl. Eng.* **2014**, 1–6. <https://doi.org/10.1155/2014/238459> (2014).
30. Guo, M. K., Qin G. S., Lu, C. & Zhao M. H. Nonlinear solution of a piezoelectric PN junction under temperature gradient. *Int. J. Appl. Mech.* <https://doi.org/10.1142/S1758825121501258> (2022).
31. Saffari, P. R., Senjuntichai, T., Saffari, P. R. & Keawsawasvong, S. Thermo-hygro-induced energy harvesting from piezoelectric sandwich cylindrical microshells with honeycomb cores. *Int. J. Mech. Mater. Des.* **21**(6), 1517–1549. <https://doi.org/10.1007/s10999-025-09792-2> (2025).
32. Dutta, R., Das, S., Ahmad, H., Meghana, A. R. & Gupta, V. Comparative analysis of double and single porosity effects on SH-wave induced vibrations in periodic porous lattices. *Soil Dyn. Earthq. Eng.* **186**, 108919 (2024).
33. Wang, H. et al. A self-powered wireless temperature sensing system using flexible thermoelectric generators under simulated thermal condition. *Measurement* **253**, 117637. <https://doi.org/10.1016/j.measurement.2025.117637> (2025).
34. Yang, W., Hao, G., Li, Z., Zhang, S. & Lu, L. Electromechanical behaviors of a PVDF beam-film coupled energy harvester under droplet impact. *PLoS ONE* **20**(4), e0319751. <https://doi.org/10.1371/journal.pone.0319751> (2025).
35. Zhong, L. et al. When nanocellulose meets liquid metal: A review of the synergistic frontier for flexible electronics. *Cellulose* **32**(17), 9787–9818. <https://doi.org/10.1007/s10570-025-06808-0> (2025).
36. Gupta, S., Dutta, R. & Das, S. Memory response in a nonlocal micropolar double porous thermoelastic medium with variable conductivity under Moore–Gibson–Thompson thermoelasticity theory. *J. Ocean Eng. Sci.* **8**(3), 263–277. <https://doi.org/10.1016/j.joes.2022.01.010> (2023).
37. Gupta, V., Barak, M. S. & Das, S. Vibrational analysis of size-dependent thermo-piezo-photo-electric semiconductor medium under memory-dependent Moore–Gibson–Thompson photo-thermoelasticity theory. *Mech. Adv. Mater. Struct.* **31**(28), 10543–10559 (2024).
38. Gupta, V., Barak, M. S., Ahmad, H., Das, S. & Almohsen, B. Response of moisture and temperature diffusivity on an orthotropic hygro-thermo-piezo-elastic medium. *J. Nonlinear Math. Phys.* **31**(1), 1–26 (2024).
39. Kumar, S., Barak, M. S., Kumari, N., Gupta, V. & Ahmad, H. The effect of viscosity and hyperbolic two-temperature on energy ratios in elastic and piezoviscothermoelastic half-spaces. *Mech. Time Depend. Mater.* **28**, 937–957 (2024).
40. Li, C., Wang, Y. & Liu, J. Stress gradient piezoelectric thermoelasticity based on nonlocal electron-phonon two-temperature thermal transport model and transient heat-shock responses analysis of multi-laminated piezoelectric composites. *Appl. Math. Model.* **143**, 116028. <https://doi.org/10.1016/j.apm.2025.116028> (2025).
41. Li, C. & Zhou, J. Space-time nonlocal multi-physics modeling of photo-carrier-thermoelastic dynamics and plasma plume absorption ultra-short pulse laser-ablated impact response in sandwich-like semiconductor composites. *Int. Commun. Heat Mass Transf.* **172**, 110445. <https://doi.org/10.1016/j.icheatmasstransfer.2025.110445> (2026).
42. Li, C., Liu, J. & He, T. Fractional-order rate-dependent thermoelastic diffusion theory based on new definitions of fractional derivatives with non-singular kernels and the associated structural transient dynamic responses analysis of sandwich-like composite laminates. *Commun. Nonlinear Sci. Numer. Simul.* **132**, 107896. <https://doi.org/10.1016/j.cnsns.2024.107896> (2024).

43. Li, C., Guo, H., He, T. & Tian, X. A rate-dependent constitutive model of piezoelectric thermoelasticity and structural thermo-electromechanical responses analysis to multilayered laminated piezoelectric smart composites. *Appl. Math. Model.* **112**, 18–46. <https://doi.org/10.1016/j.apm.2022.07.025> (2022).
44. Fekry, M. Non-local thermal stresses on fiber-reinforced viscoelastic porous medium. *Continuum Mech. Thermodyn.* **38**(1), 4. <https://doi.org/10.1007/s00161-025-01429-1> (2026).
45. Nandi, S. & Sarkar, S. P. A fractional three-phase-lag approach incorporating Atangana–Baleanu derivative for thermoelastic analysis of porous solids. *J. Vib. Eng. Technol.* **14**(2), 89. <https://doi.org/10.1007/s42417-025-02277-8> (2026).
46. Lazar, M. & Agiasofitou, E. Nonlocal elasticity of Klein–Gordon type: Fundamentals and wave propagation. *Wave Motion* **114**, 103038. <https://doi.org/10.1016/j.wavemoti.2022.103038> (2022).
47. Agiasofitou, E. & Lazar, M. Nonlocal elasticity of Klein–Gordon type with internal length and time scales: Constitutive modelling and dispersion relations. *PAMM*. <https://doi.org/10.1002/pamm.202300065> (2023).
48. Eringen, A. C. On differential equations of nonlocal elasticity and solutions of screw dislocation and surface waves. *J. Appl. Phys.* **54**(9), 4703–4710. <https://doi.org/10.1063/1.332803> (1983).
49. Gupta, V., Barak, M. S. & Das, S. Impact of memory-dependent heat transfer on Rayleigh waves propagation in nonlocal piezo-thermo-elastic medium with voids. *Int. J. Numer. Methods Heat Fluid Flow* <https://doi.org/10.1108/HFF-10-2023-0615> (2024).
50. Iovane, G. & Nasedkin, A. V. Modal analysis of piezoelectric bodies with voids. II. Finite element simulation. *Appl. Math. Model.* **34**(1), 47–59. <https://doi.org/10.1016/j.apm.2009.03.02> (2010).
51. Abouelregal, A. E. Three-phase-lag thermoelastic heat conduction model with higher-order time-fractional derivatives. *Indian J. Phys.* **94**(12), 1949–1963. <https://doi.org/10.1007/s12648-019-01635-z> (2020).
52. Biot, M. A. Thermoelasticity and irreversible thermodynamics. *J. Appl. Phys.* **27**(3), 240–253. <https://doi.org/10.1063/1.1722351> (1956).
53. Lord, H. W. & Shulman, Y. A generalized dynamical theory of thermoelasticity. *J. Mech. Phys. Solids* **15**(5), 299–309. [https://doi.org/10.1016/0022-5096\(67\)90024-5](https://doi.org/10.1016/0022-5096(67)90024-5) (1967).
54. Tzou, D. Y. A unified field approach for heat conduction from macro- to micro-scales. *J. Heat Transf.* **117**, 8–16. <https://doi.org/10.1115/1.2822329> (1995).
55. Choudhuri, S. R. On a thermoelastic three-phase-lag model. *J. Therm. Stress.* **30**(3), 231–238 (2007).
56. Sharma, J. N., Walia, V. & Gupta, S. K. Effect of rotation and thermal relaxation on Rayleigh waves in piezothermoelastic half space. *Int. J. Mech. Sci.* **50**(3), 433–444. <https://doi.org/10.1016/j.ijmecsci.2007.10.001> (2008).
57. Sharma, M. D. Propagation of generalised Rayleigh wave at the surface of piezoelectric medium with arbitrary anisotropy. *ZAMM* **103**(3), 1–15. <https://doi.org/10.1002/zamm.202000160> (2023).
58. Gupta, V. & Barak, M. S. Fractional and MDD analysis of piezo-photo-thermo-elastic waves in semiconductor medium. *Mech. Adv. Mater. Struct.* <https://doi.org/10.1080/15376494.2023.2238201> (2023).
59. Chiriță, S., Ciarletta, M. & Tibullo, V. On the thermomechanical consistency of the time differential dual-phase-lag models of heat conduction. *Int. J. Heat Mass Transf.* **114**, 277–285. <https://doi.org/10.1016/j.ijheatmasstransfer.2017.06.071> (2017).
60. Zampoli, V. On the increase in signal depth due to high-order effects in micro- and nanosized deformable conductors. *Math. Probl. Eng.* **2019**, 1–11. <https://doi.org/10.1155/2019/2629012> (2019).
61. Quintanilla, R. & Racke, R. A note on stability in three-phase-lag heat conduction. *Int. J. Heat Mass Transf.* **51**(1–2), 24–29. <https://doi.org/10.1016/j.ijheatmasstransfer.2007.04.045> (2008).
62. Gupta, V., Ahmad, H., Barak, M. S., Das, S. & Kumar, S. Rayleigh wave in nonlocal piezo-thermo-electric semiconductor medium with fractional MGT model. *Waves Random Complex Med.* 1–24 (2024).
63. Eringen, A. & Wegner, J. Nonlocal continuum field theories. *Appl. Mech. Rev.* **56**(2), B20–B22. <https://doi.org/10.1115/1.1553434> (2003).
64. Kumar, M., Kaswan, P., Sarkar, N., Liu, X. & Kumari, M. Rayleigh waves in nonlocal generalized thermoelastic media. *Int. J. Numer. Methods Heat Fluid Flow* **33**(6), 2049–2072. <https://doi.org/10.1108/HFF-08-2022-0468> (2023).

Acknowledgements

The authors gratefully acknowledge Sri Mata Amritanandamayi Devi (Amma), Chancellor, Amrita Vishwa Vidyapeetham, for her inspiration and for providing financial support for the Article Processing Charges (APC) of this publication.

Author contributions

Mohammed Aldandani: Supervision. Vipin Gupta: Methodology, Software. Arun Kumar Yadav: Conceptualization, Investigation. Abhik Sur: Visualization, Writing – review & editing. Murat Yaylacı: Formal analysis, Validation, Resources. Soumik Das: Visualization, Software, Writing – original draft.

Funding

Open access funding provided by Amrita Vishwa Vidyapeetham.

Declarations

Competing interests

The authors declare no competing interests.

Data availability

All data generated or analyzed during this study are included in this published article.

Additional information

Correspondence and requests for materials should be addressed to S.D.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026